FAIRagro

Data Reuse

What does reusing data really mean?

Using data for purposes other than those for which it was originally collected for.

Some benefits:

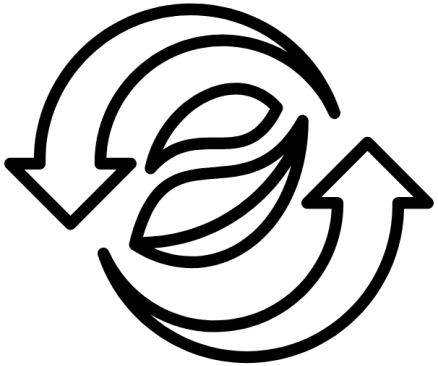
- ➔ obtain **reference data** for your research
- ➔ **avoid doing new** or unnecessary experiments
- ➔ **aggregating results** obtained from different methods or samples
- ➔ **connecting** and meta-analysing datasets for novel insights



Created by Dewi Novita Sari
from Noun Project

Illustration: “definition” by Dewi Novita Sari is licensed under CC BY 3.0

Difficulties in reusing agricultural research data



Created by littlestar23
from Noun Project

- **Data complexity**.
- **Data quality** standards differ across disciplines.
- **Data handling** since size and number of datasets rises.

➔ Datasets are useful for original producer, but reuse ist difficult

- **Epistemologically problems:** ambiguities, missing documentation, …
- **Logistic problems:** undiscoverable data, different data types, …

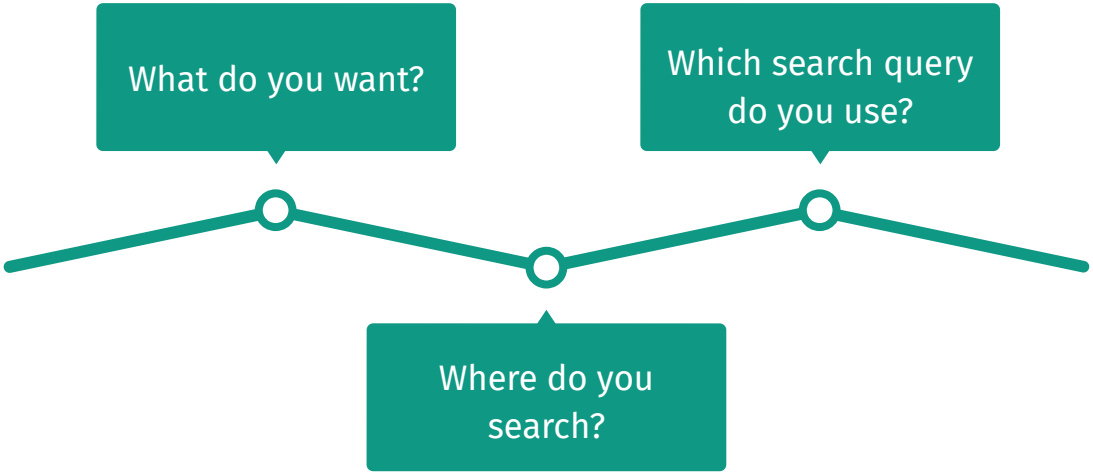
Source: modified from Papoutsoglou et al. 2023; Illustration: “reusable” by littlestar23 from Noun Project (CC BY 3.0)

FAIRagro

Finding Data

for Reuse

Finding Data of interest





Created by mangunkarsa from Noun Project

Source: modified from “Finding Data” by The Turing Way Community; “Research” by mangunkarsa is licensed under CC BY 3.0

What Data you need and why you need them?

What do you want?

Before starting to search for data

- ➔ how will you use the desired data in the context of your overall research question?
- ➔ for comparison or validation?
- ➔ as basis for a new study?

Characteristics of the required data:

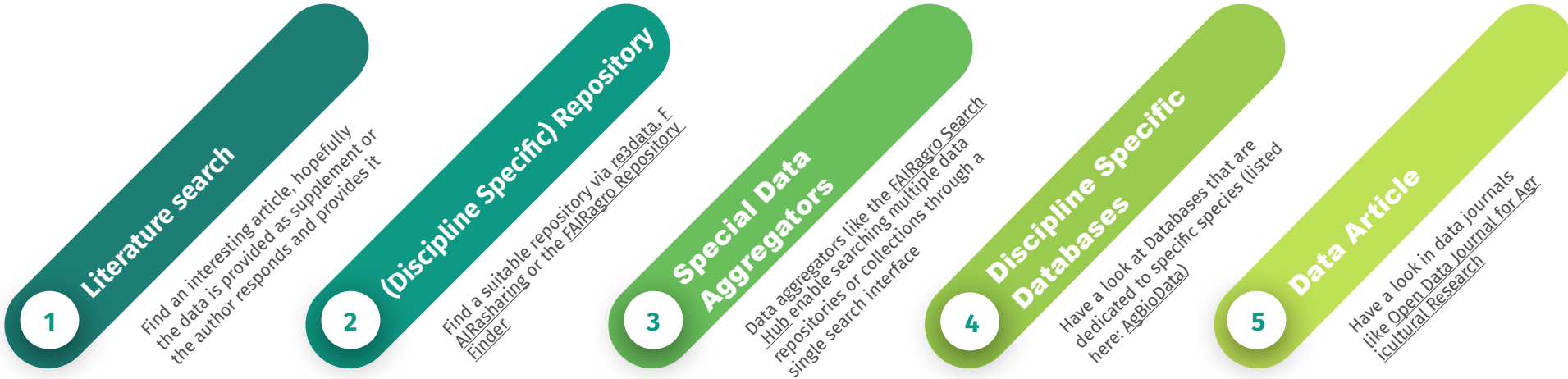
- ❑ data format?
- ❑ spatial or temporal coverage?
- ❑ availability?
- ❑ author or research group?



Source: modified from Gregory et al. (2018); “pen” by ArashDesign is licensed under CC BY 3.0

Select the most appropriate resource

Where do you search?

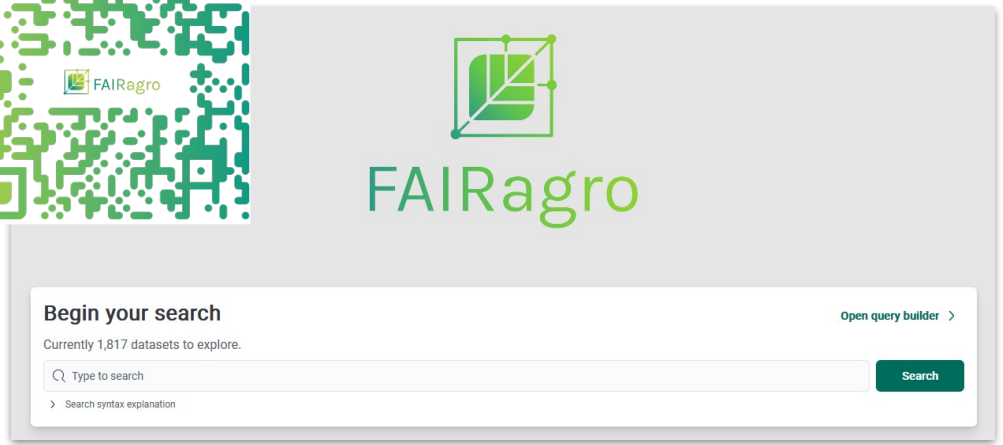


Source: modified from Gregory et al. (2018), "Finding Data" by The Turing Way Community

Data aggregators

Domain-agnostic data search:

- B2FIND data search
- BASE
- DataCite Metadata Search
- Europäisches Datenportal
- Google Dataset Search
- Data Catalogue at EOSC Portal



Life Science data search:

- LIVIVO - ZB MED Search Portal for Life Sciences

Agricultural Science data search:

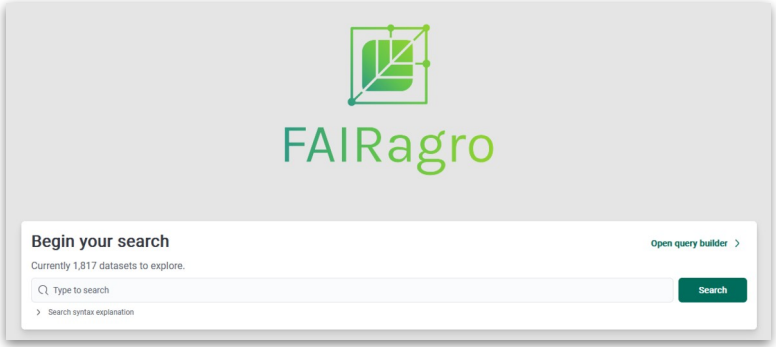
- FAIRagro Search Hub

Source: modified from "Finding Data" by The Turing Way Community; FAIRagro Search Hub

FAIRagro Search Hub

Searching multiple agricultural repositories at a time!





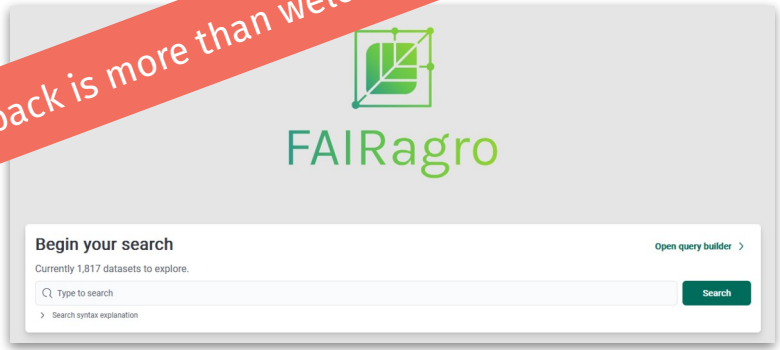
various search and **filter functions**, e.g. based on **crop species** or **soil type**

Source: FAIRagro Search Hub

FAIRagro Search Hub

Searching multiple agricultural repositories at a time!

Your feedback is more than welcome!



various search and **filter functions**, e.g. based on **crop species** or **soil type**

Source: FAIRagro Search Hub

How to search for data?

Which search query do you use?

Describing your desired data **effectively** is **key** to communicating with the search system.

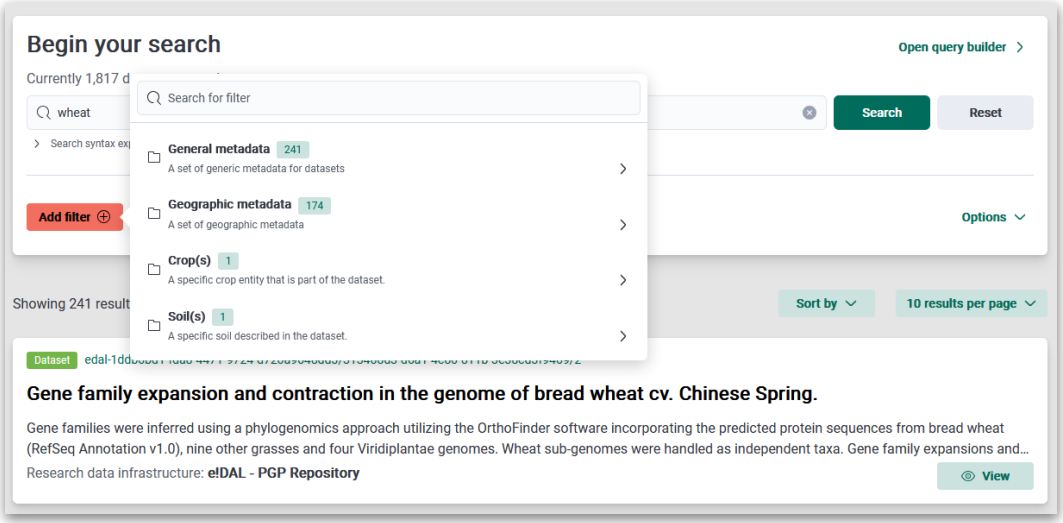
Your description **will determine if relevant data are retrieved** and may inform the order of the hits in the results list.

Source: modified from Gregory et al. (2018)

Step 1: use the filters

Which search query do you use?

➔ Use filters to refine your search

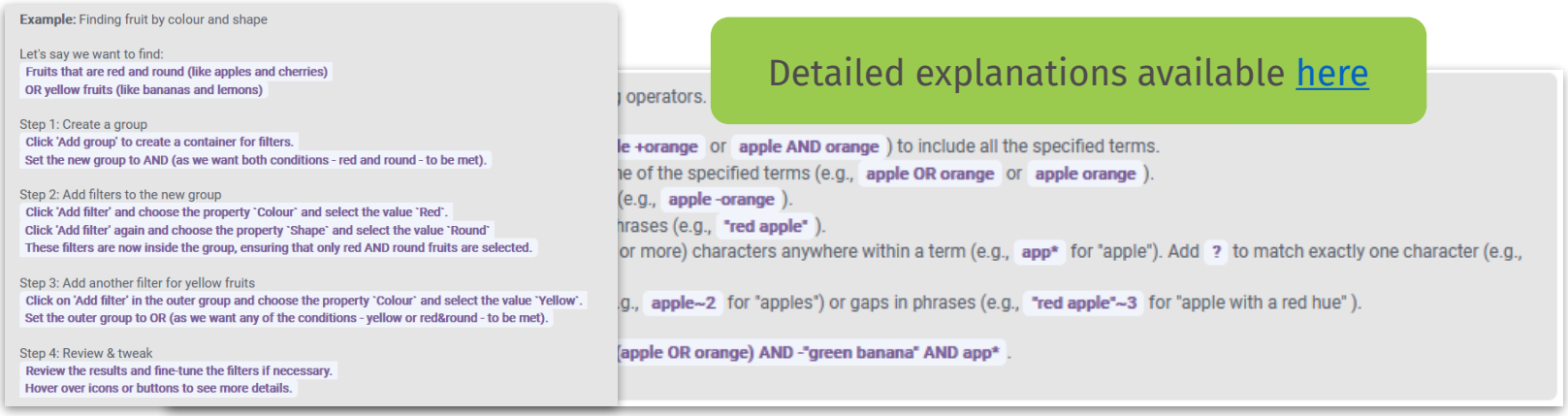


Source: Screenshot from FAIRagro Search Hub taken on 24.07.25

Step 2: build your query

Which search query do you use?

➔ Use operators to build your query



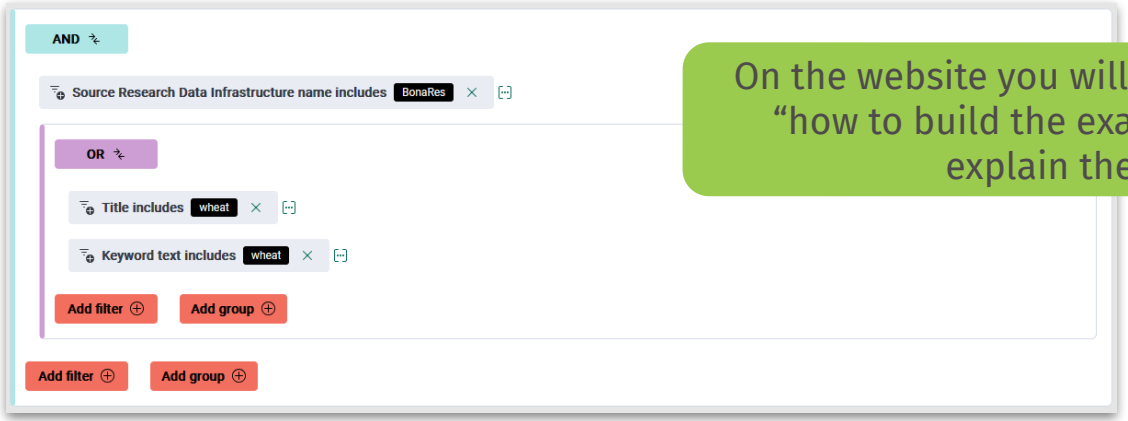
Detailed explanations available here

Source: Screenshot from FAIRagro Search Hub taken on 24.07.25

Step 3: use the query builder

Which search query do you use?

➔ The query builder may help you!



On the website you will find a pre-build example, a “how to build the example” tour and a tutorial explain the query builder!

Source: Screenshot from FAIRagro Search Hub taken on 24.07.25

Step 4: refine your search

- Repositories and Data Aggregators use different strategies to make the data more discoverable
  - familiarize yourself with the controlled vocabularies, subject categories and search field of your resource
- **Refine** your search after first try
  - broaden or narrow your approach based on the retrieved results
- Explore interesting looking results
  - these can link to related ones (e.g., from the data provider, from different time series, ...)

Source: modified from Gregory et al. (2018)

FAIRagro

Check Reusability

Check the license

Am I allowed to use that data?

Am I allowed to alter it/combine it with other data?



Created by iconisme
from Noun Project

Illustration: “question” by iconisme from the Noun Project is licensed under CC BY 3.0

CC copyright licenses: five components



Attribution (BY): author/creator must always be credited

Non-Commercial (NC): must not be used for commercial purposes

Share Alike (SA): redistribution/adaptations only under the same conditions

No Derivatives (ND): no adaptations or derivatives are permitted

Public domain: none of the above limitations

Source: Bobrov *et al.* 2021, Cozatl *et al.* 2021 and Rathmann *et al.* 2021 modified by ZB MED 2022

Exercise: Combining Licenses

You would like to combine datasets with different licenses. What is the resulting license?

1. 

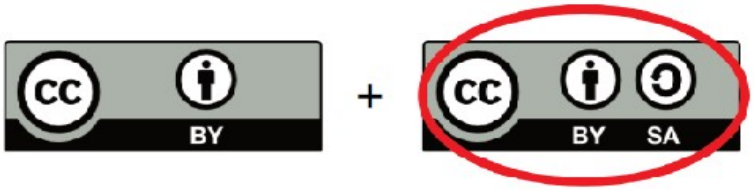
+ 

= ?

Source: Bobrov et al. 2021

Exercise: Combining Licenses - Solution

You would like to combine datasets with different licenses. What is the resulting license?

1. 

Source: Bobrov et al. 2021

Exercise: Combining Licenses

You would like to combine datasets with different licenses. What is the resulting license?

2. 

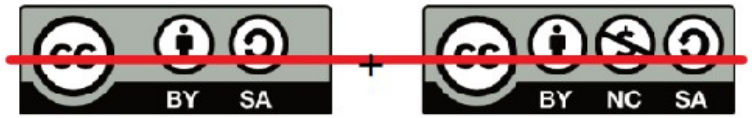
+ 

= ?

Source: Bobrov et al. 2021

Exercise: Combining Licenses - Solution

You would like to combine datasets with different licenses. What is the resulting license?

2. 

Source: Bobrov et al. 2021

Exercise: Combining Licenses

You would like to combine datasets with different licenses. What is the resulting license?

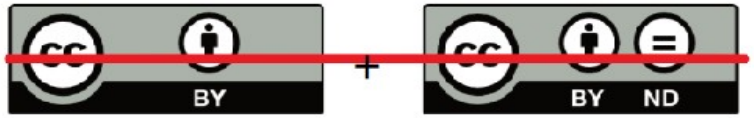
3. 

+ = ?

Source: Bobrov et al. 2021

Exercise: Combining Licenses - Solution

You would like to combine datasets with different licenses. What is the resulting license?

3. 

Source: Bobrov et al. 2021

Check the license

Am I allowed to use that data?

Am I allowed to alter it/combine it with other data?

Am I allowed to republish it?

Do I need to name the author?



Created by iconisme from Noun Project

Illustration: “question” by iconisme from the Noun Project is licensed under CC BY 3.0

Give back!

1 **Cite** the data properly:

- ➔ Following the **good scientific practice**, you **must cite** sources regardless of license or copyright protection!
- ➔ Data availability statement examples:
  - **Using the Digital Object Identifier (DOI):**
    "The data that support the findings of this study are openly available in [repository name] at http://doi.org/[doi]."
  - **If no DOI is issued:**
    "The data that support the findings of this study are openly available in [repository name] at [URL], reference number [reference number]."

Source: modified from Gregory et al. (2018), "Finding Data" by The Turing Way Community

Give back!

2. **Provide feedback** to data creators or the data repository about any issues associated with data accessibility, data quality, interoperability or metadata completeness

3. **Share** your own data!

Source: modified from Gregory et al. (2018), "Finding Data" by The Turing Way Community